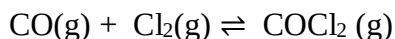


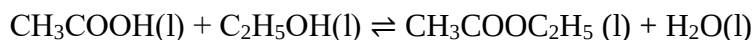
TUTORIAL CHAPTER 3 (i)

1. Carbonyl chloride (COCl_2) also called phosgene, was used in World War I was a poisonous gas. The equilibrium concentrations for the reaction between carbon monoxide (CO) and molecular chlorine (Cl_2) to form COCl_2 is given below :

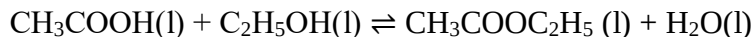


At equilibrium with 77°C , there are $[\text{CO}] = 2.1 \times 10^{-3} \text{ M}$, $[\text{Cl}_2] = 0.065 \text{ M}$, and $[\text{COCl}_2] = 0.42 \text{ M}$. Calculate the equilibrium constant, K_c

2. Nitrogen reacts with Chlorine to produce Nitrogen Trichloride. At equilibrium, the concentrations of each gas were found to be $[\text{N}_2] = 0.15 \text{ M}$, $[\text{Cl}_2] = 0.25 \text{ M}$, and $[\text{NCl}_3] = 0.50 \text{ M}$. Calculate the value of the equilibrium constant K_c .
3. Sulfur Dioxide reacts with Oxygen gas to produce Sulfur Trioxide. At equilibrium, the partial pressures of each gas were found to be 0.10 atm , 0.30 atm , and 0.45 atm for SO_2 , O_2 , and SO_3 respectively. Calculate the equilibrium constant K_p .
4. In a 250 mL solution, 1 mol of glacial ethanoic acid is mixed with 1 mol of absolute ethanol in a closed flask and allowed to achieve equilibrium at room temperature. The equilibrium mixture was found to contain 0.67 mol of ethyl ethanoate. Calculate K_c for the following equilibrium at room temperature.

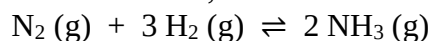


5. The equilibrium constant for the following reaction is 4.12 at 298 K ,



Calculate the mass of ethyl ethanoate produced from a mixture containing 2 mol of ethanoic acid and 2 mol of ethanol in the 250 mL solution.

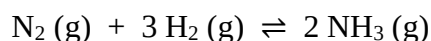
6. If K_c for the reaction shown below is 9.60 at 300°C , what is the value of K_p at this temperature? (b) If $K_p = 145 \times 10^{-3}$ at 500°C , what is the value of K_c at this temperature?



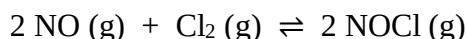
7. The equilibrium constant, K_p for the decomposition of carbonic acid (H_2CO_3) in soft drinks is found to be 1.05 at 250°C . If the equilibrium partial pressure of H_2CO_3 and CO_2 are 0.875 atm and 0.463 atm, respectively, what is the equilibrium partial pressure of H_2O at 250°C ?



8. At the start of a reaction, there a 0.429 mol N_2 , 0.0123mol H_2 , and 6.24×10^{-3} mol NH_3 in a 5.30 L reaction vessel at 375°C . If the equilibrium constant, K_c for the reaction is 1.2 at this temperature, decide whether the system is at equilibrium. If it's not, predict which way the equilibrium will proceed.



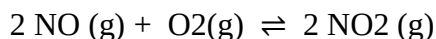
9. Nitrosyl chloride, NOCl is produced from the reaction between nitric oxide gas and chlorine gas. At temperature $T^\circ\text{C}$, the partial pressures of NO , Cl_2 and NOCl in a 1 dm^3 equilibrium mixture are 60 kPa, 50 kPa and 80 kPa respectively.



- Calculate the value of equilibrium constant, K_p .
 - When additional Cl_2 is added, its new partial pressure is 70 kPa. The new equilibrium partial pressure of NOCl is 88 kPa. Determine the partial pressure of NO and Cl_2 in the new equilibrium mixture if the temperature remains constant.
10. At 445K, the equilibrium constant for the breakdown of NO_2 is 1.8×10^{-6} . Suppose 0.5 mol of NO_2 is placed in a 1.00 dm^3 reaction vessel. Calculate the concentration of each species at equilibrium.

The equation for the reaction is : $2 \text{NO}_2(\text{g}) \rightleftharpoons 2 \text{NO}(\text{g}) + \text{O}_2(\text{g})$

11. A natural source of nitrogen oxides (NO) occurs from a lightning stroke. The very high temperature in the vicinity of a lightning bolt causes the gases oxygen and nitrogen in the air to react to form nitrogen oxide. The nitrogen oxide then reacts very quickly with more oxygen to form nitrogen dioxide as shown below :



At equilibrium with 90°C , there are $[\text{NO}] = 13.6 \times 10^{-2} \text{ M}$, $[\text{O}_2] = 0.54 \text{ M}$, and $[\text{NO}_2] = 0.41 \text{ M}$. Calculate the equilibrium constant, K_c and then K_p .



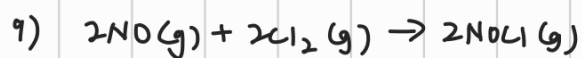
The equilibrium constant, K_c for the above reaction at 200°C is 20.0. At the point during the reaction, a mixture containing 1.0 mol dm^{-3} of H_2 , 1.3 mol dm^{-3} of I_2 and 1.5 mol dm^{-3} of HI is left standing at 200°C .

- (i) Write the reactions quotient, Q_c for the reaction.
- (ii) Calculate the reaction quotient, Q_c for the reaction.
- (iii) Determine whether the reaction is in the equilibrium or not. Then, determine the net direction the reaction is progressing.

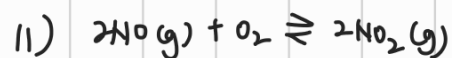
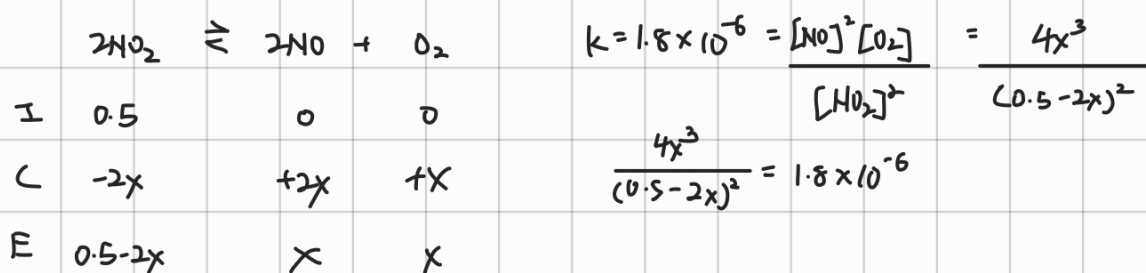
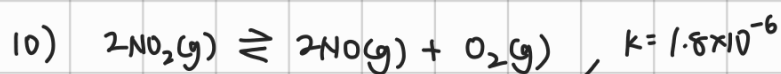
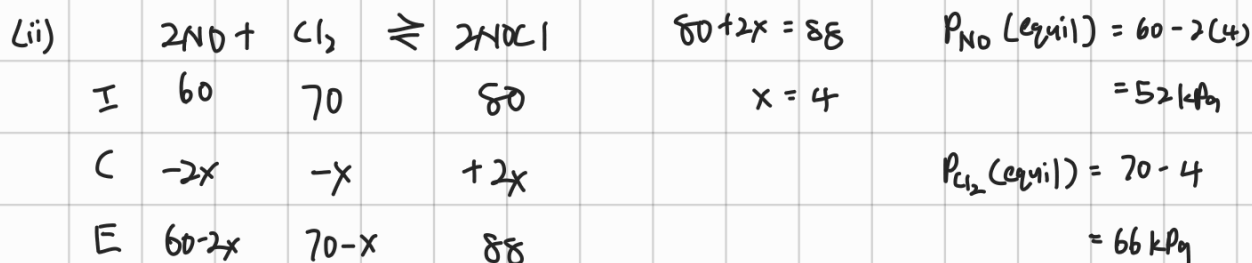
13. The equilibrium constant, K_p for the decomposition of phosphorus pentachloride gas to phosphorus trichloride gas and chlorine gas is 1.00 at 250°C .

- (i) Write down the balanced chemical equation including their physical state and the equilibrium constant in terms of K_p for this reaction.
- (ii) Calculate the partial pressures of all reacting species at the equilibrium if the initial pressure partial pressure of phosphorus pentachloride is 0.25 atm flask at 250°C .

14. Aurum (III) chloride, (AuCl_3), has a solubility of 54 g/100 mL of water. Determine the solubility-product constant, K_{sp} , of AuCl_3 in water.

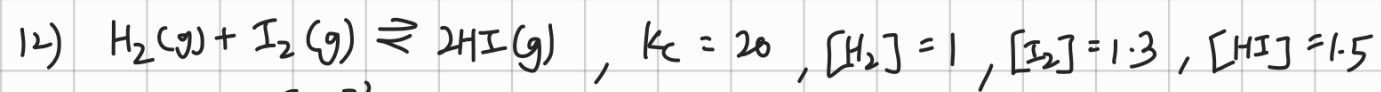


$$\begin{aligned} \text{ci)} \quad K_p &= \frac{(P_{\text{NOCl}})^2}{(P_{\text{NO}})^2 (P_{\text{Cl}_2})} \\ &= \frac{80000^2}{60000^2 (50000)} \\ &= 3.56 \times 10^{-5} \end{aligned}$$



$$K_c = \frac{[\text{NO}_2]^2}{[\text{NO}]^2 [\text{O}_2]} = \frac{(0.41)^2}{(13.6 \times 10^{-2})^2 (0.54)^2} = 31.17$$

$$\begin{aligned} K_p &= K_c \times R T^{\Delta n} \\ &= 31.17 (0.0821 \times 363)^{2-3} \\ &= 1.05 \end{aligned}$$



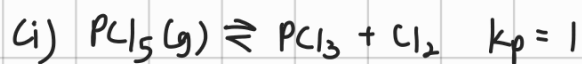
i) $Q_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]}$

ii) $Q_c = \frac{1.5^2}{1 \times 1.3} = 1.731$

cii) $Q_c = 1.731$, $K_c = 20.0$

$Q_c < K_c$, reaction shift to right, reaction is not equilibrium

13) $K_p = 1$



$$1 = \frac{x^2}{0.25-x}$$

$$0.25 - x = x^2$$

$$x^2 + x - 0.25 = 0$$

$$4x^2 + 4x - 1 = 0, x = 0.21$$

14) solubility = molar mass $\times$ molar solubility, m.m of $\text{AlCl}_3 = 303$

$$0.54 = 303 \cdot x, x = 0.0018 \text{ mol L}^{-1}$$

$$K_{sp} = [\text{Al}^{3+}][\text{Cl}^-]^3 = x(3x)^3 = 27x^4$$

$$= 27(0.0018)^4$$

$$= 2.83 \times 10^{-10} \text{ M}^4$$